

Gun launch to orbit on cold hydrogen

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1 Introduction

A kilogram in low orbit carries about 9 kWh of kinetic energy, which is roughly fifty cents of electricity at a 6 ¢/kWh industrial rate. A gun fixed to the ground keeps all of its machinery on the ground, so its recurring launch cost can in principle approach that energy bill; the drive developed here brings the principle within reach, converting over half of its delivered gas energy into projectile motion where a laboratory light-gas gun converts a few percent of its propellant energy. The obstacle is the drive: accelerating a projectile to 8 km/s by gas pressure on its base requires hydrogen far hotter than a reusable steel barrel tolerates, and, as shown below, stalls short of 8 km/s at any temperature. This paper develops an alternative: cold hydrogen injected radially through wall ports onto a slender cone trailing the projectile. The sections below show, by flow simulation and a system ledger, that this drive reaches 8 km/s on 550 K gas, that over half of the delivered jet energy reappears as projectile kinetic energy, that the gas system needs less pressure than a classical gun needs mean base pressure, and that at the two design points which emerge from the results the electricity bill is \$3.80 and \$2.50 per kilogram of payload. The launch-cost problem then sits in vehicle hardware, not in the launcher.

2 Cold gas on a slender cone reaches 8 km/s; hot columns do not

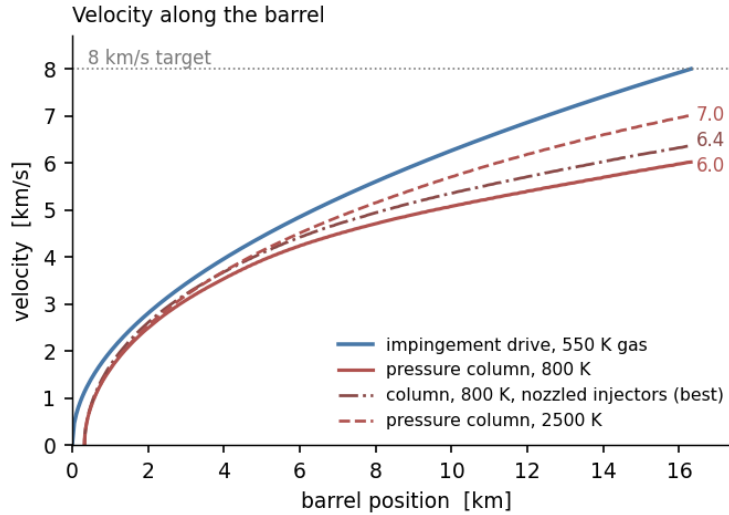


Figure 1: Velocity along a 16.3km barrel from the unsteady interior-ballistics model: the constant-pressure column at two reservoir temperatures, the best nozzled-injection variant from the injection-velocity sweep, and the impingement drive at its constant design acceleration.

Two gas temperatures appear throughout. At 800 K and below, every wetted part of the gas system can be plain pressure-vessel steel, uncooled; above it, the materials change kind, and 2500 K stands in for what refractory linings and cooled components could reach.

A single breech reservoir cannot drive a barrel longer than a few hundred meters: the gas would expand thousands-fold on the way down the tube, so a useful end pressure demands an impossible starting one.

The pressure behind the projectile must instead be renewed by injection along the barrel, and the natural scheme is a distributed column: ports add gas behind the base, and the column pushes. The flaw is that injected gas arrives with no motion along the bore, so the column itself must accelerate every fresh kilogram to the projectile's speed before that kilogram can push, an internal cost that grows with the square of speed while the push it delivers does not. Nozzling the injectors so the gas arrives already moving along the bore does not remove this cost, because the nozzle draws the jet velocity from the gas's own heat, and the budget is small: expanding 800 K hydrogen through a nozzle yields at most $\sqrt{2c_p T} \approx 4.8$ km/s, and gas near that limit arrives with almost no pressure left to push with. A sweep of the injection velocity confirms it: the optimum, at 1.8 km/s nozzles, gains 0.3 km/s over plain radial feed while demanding eight times the supply pressure. The model (Methods) gives the consequence: at 800 K the column stalls at 6.0 km/s plain and 6.4 km/s nozzled, and raising the reservoir to 2500 K reaches only 7.0; at 3500 K, 7.2. Hotter gas carries more energy, but the same architecture consumes the increase accelerating its own working fluid. The failure is systemic, not thermal.

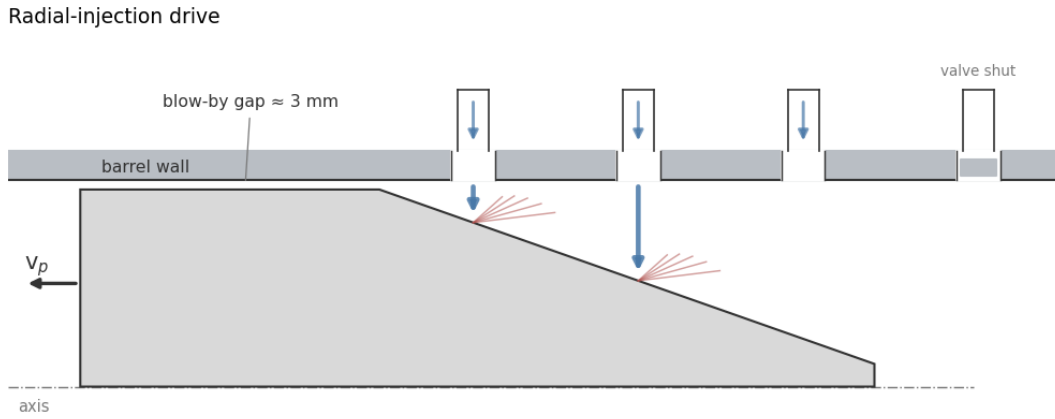


Figure 2: The drive in half-section: wall ports open as the projectile's shoulder passes and blow hydrogen radially onto the receding boat-tail cone.

The alternative is to stop maintaining a pressure column and let injected gas push on the projectile directly. The projectile trails a slender cone; the barrel wall carries rings of ports; the gas enters radially, so it never needs bore-direction momentum, and meets a surface receding from it at a shallow angle. Each port is sealed by a burst disk, a thin metal membrane ruptured on command by an exploding-foil initiator (a detonator in which a capacitor discharge vaporizes a foil), which opens the port in microseconds, faster than any mechanical valve, just as the shoulder arrives. Seen from the projectile, the port flow comes in as an oblique stream, and its expansion against the retreating cone surface transfers pressure work to the vehicle: a gas piston whose cylinder is the cone, re-created at each port for the milliseconds the cone takes to pass. No column of moving hydrogen ever forms, because each port's gas does its work within a cone-length of where it entered. The obvious objection is that a 1500 m/s jet cannot service a body at 8000 m/s, since it can never catch it. The cone geometry removes the need to: the port fires as the shoulder passes overhead, the receding cone surface stays within reach for the whole pass, and the gas finishes its work before the surface is gone. The energy budget is then set not by the jet's axial speed but by the total thermal-plus-kinetic content of the gas, and of that budget the shot-average conversion into projectile energy exceeds one half (Section 5); a laboratory two-stage light-gas gun, the standard instrument at these speeds, delivers a few percent of its propellant energy, with the rest lost to its piston, gas heating, and friction. The gas is supplied at 550 K, which keeps every wetted wall below 600 K; that limit is imposed so the whole barrel can be ordinary pressure-vessel steel, run without active cooling and without exotic alloys or liners (assumed throughout). The push is smooth: the simulated single-port force shows no overshoot and a settled ripple below 0.1%, so peak acceleration exceeds the mean only by the 3–10% unevenness of port-to-port handover.

3 The drive physics scales across bore diameters

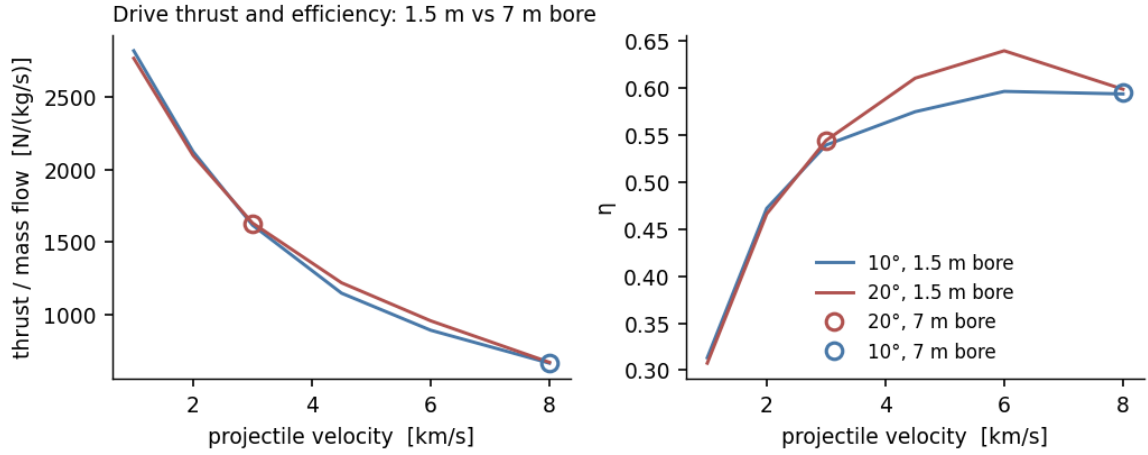


Figure 3: Thrust per unit gas flow and conversion efficiency against projectile velocity: curves from simulations at 1.5 m bore, circles from simulations at 7 m bore with all port hardware scaled in proportion and the clearance gap held fixed.

At these speeds the friction forces in the gas are roughly a thousand times smaller than the pressure forces, and a flow governed by pressure alone has no length scale of its own: enlarge every dimension by the same factor and the pressures and velocities reproduce exactly. The one part that does *not* scale is the clearance gap: the projectile must run the barrel without touching the wall, since metal contact at kilometers per second would destroy both, so a few millimeters of radial clearance are left and some gas leaks forward past the body. That gap is set by manufacturing tolerance, not aerodynamics, and stays a few millimeters at any bore. Ahead of the vehicle the barrel is pumped down to about 100 Pa before the shot, a routine industrial vacuum, behind a light muzzle diaphragm the vehicle bursts on exit; the residual air over the full length then totals tens of kilograms against a multi-tonne projectile, half a percent of its mass. Gas leaking forward through the gap during the shot accumulates ahead of the vehicle as well; the interior model charges leakage at under two percent of muzzle velocity for gaps up to several millimeters (derived). Two bore sizes are used in this paper, 1.5 m and 7 m, and the drive simulations were run at both, with every port dimension scaled in proportion and the gap held at 3 mm: thrust and efficiency reproduce to -0.1% and $+0.3\%$. One simulated map therefore serves both designs, and the large one inherits physics checked at the small scale rather than extrapolated to it.

4 Thrust falls with speed

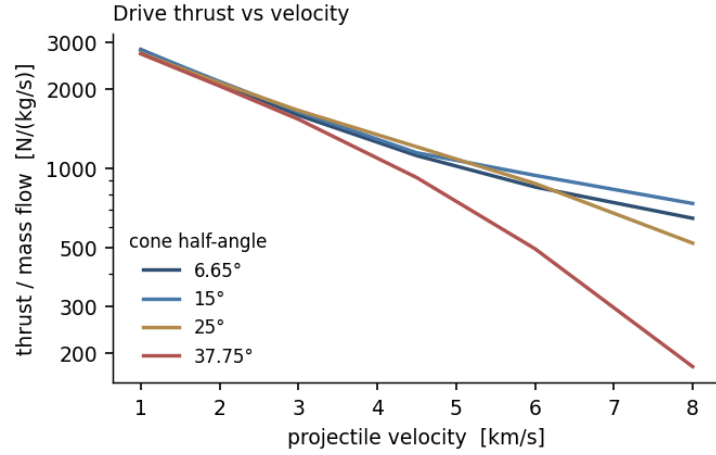


Figure 4: Thrust per unit of gas flow against projectile velocity from the 65-case simulation campaign at fixed port conditions, one curve per cone half-angle.

The quantity that sizes the whole gas system is the thrust obtained per kilogram-per-second of injected hydrogen. It falls with speed for every geometry, from about 2800 N per kg/s at 1 km/s to 650–740 at 8 km/s for the slender cones, because the pressure the gas can exert drops as the surface recedes faster. On steep cones it does worse than fall: at 37.75° the thrust collapses above 6 km/s, because a steep cone is axially short and at high speed the gas has not finished expanding when the surface ends.

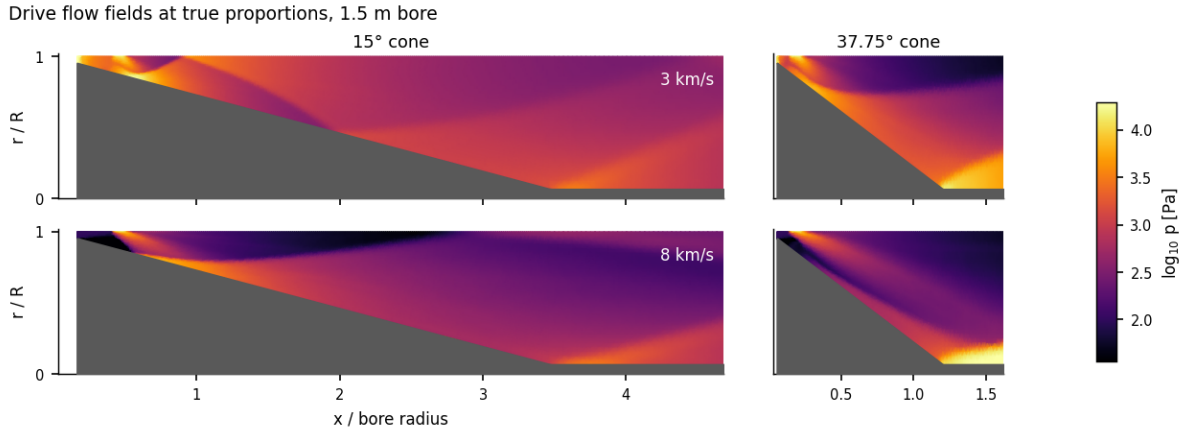


Figure 5: Pressure fields in the drive cavity drawn at true geometric proportions: columns compare a slender (15°) and the steepest (37.75°) cone, rows compare mid and full speed, on a common color scale.

The fields show both regimes. On the slender cone the port stream spreads into an expansion that stays pressed against the surface at every speed. On the steep cone at 8 km/s most of the pressure appears beyond the cone’s end, where it can no longer act on the vehicle. The same fields computed at 7 m bore are indistinguishable from these once normalized.

5 Slender cones give the highest efficiency and the lowest pressure

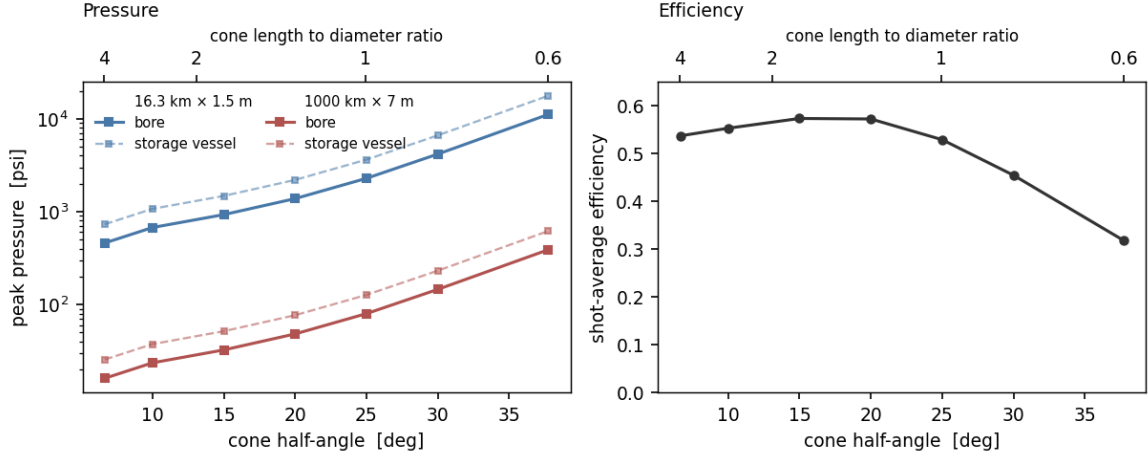


Figure 6: Against cone half-angle: peak drive pressure (left panel, logarithmic; solid for the pressure one bore cross-section must be sized for, faded dashed for the storage vessel) and the shot-average conversion efficiency, which is bore-independent (right panel). Top axes give the equivalent cone length in bore diameters.

One limit sits outside the model and is stated first: the cone is structure that must be carried and flown, and neither its mass nor its effect on the vehicle’s aerodynamic balance is included here. A long cone adds tail mass and tail surface, both of which move the vehicle’s balance rearward, so slenderness cannot be pushed arbitrarily far; the length limit of four bore diameters used below stands in for that constraint qualitatively.

Within that limit the trade is clean. The shot-average efficiency, projectile kinetic energy over total delivered jet energy, follows from integrating the thrust map over the shot: it peaks at 0.57 for cones of 15–20° and stays above 0.53 out to the slenderness limit, while the steepest cone manages 0.32 (more detail in Methods). Pressure discriminates more sharply than efficiency. A slender cone is long, many ports work on it at once, and the thrust demand divides among them; a steep cone concentrates the same demand on a few ports, whose pressure rises steeply. At the slenderness limit (6.65° half-angle) the 1.5 m gun’s bore must be sized for 460 psi at the working section and its storage vessels for 730 psi; the 7 m gun needs 17 and 26 psi. For comparison, a classical gun of the same length and payload would need a mean base pressure of 1050 and 37 psi respectively: this drive undercuts even that, and for a plain reason. A classical gun must hold its mean pressure across the entire bore cross-section continuously from the breech, while here the force is produced locally by many small jets, so pipes and vessels hold modest pressure and the full working pressure exists only inside the jets. At a few hundred psi the gas system operates at ordinary pipeline pressure, below the regulatory boundary above which vessels must be forged rather than welded, and the barrel stays ordinary welded steel, which keeps its capital cost low. The hydrogen itself is recovered after each shot: a muzzle suppressor of the same baffled-chamber construction as a shotgun silencer, with a steel ball in its first chamber that seats against the bore opening immediately after the projectile passes, closes the barrel and keeps most of the gas inside for pump-back.

6 Two designs emerge from the maps

Table 1: Two working points read off the maps, sized by acceleration tolerance.

	demo cargo	crewed vision
bore / barrel	1.5 m / 16.31 km	7 m / 1000 km
peak acceleration	200 g	6 g
launch mass / payload	6.49 t / 1.28 t	303 t / 91.3 t
gas supply pressure	730 psi	26 psi
electricity per shot	79 MWh	3.7 GWh
electricity cost (6 ¢/kWh)	\$3.8 / kg	\$2.5 / kg

The drive physics enforces no size; the payload’s acceleration tolerance does, through barrel length $L = v^2/2a$. The smaller design answers the question: what is the least gun that delivers below fifty dollars per kilogram, the level at which orbital energy infrastructure such as space-based solar power becomes economic to deploy in quantity? Rugged cargo tolerates 200 g , which sets 16.3 km of barrel; a 1.5 m bore then closes the vehicle around 1.28 t of payload. The larger answers: what carries people and lands below ten dollars per kilogram, the level needed to move industry and settlement off Earth? The ceiling for a trained human over the required minutes is 6 g , which the formula alone would turn into 544 km of barrel; but the design does not run the gun at the ceiling. The ride’s largest loads come after the muzzle, where drag and the pitch-over turn reach 4.5 g under the trim schedule, so the gun phase is held to a 3.3 g mean and the barrel stretches to 1000 km at 7 m bore and 91 t of payload, keeping the entire ride under the ceiling with margin. These are working points, not commitments; every gun between them runs on the same maps, the same cone, the same port conditions, and the same 550 K hydrogen.

7 The vehicle flies stable at Mach 23.5

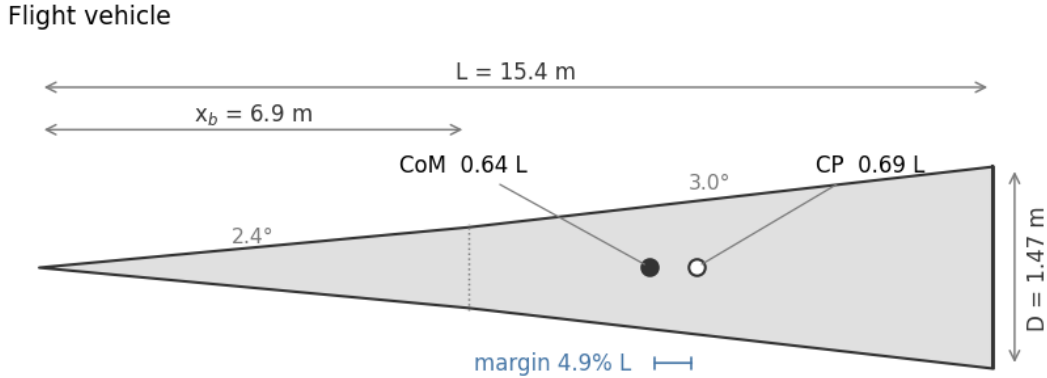


Figure 7: The flight vehicle of the smaller design: a two-cone body with the mass centre (filled), pressure centre (open), and their separation marked.

The vehicle leaves the muzzle at Mach 23.5 in sea-level air, and its trajectory, dispersion, and structural loads all rest on two aerodynamic derivatives. C_N is the normal-force coefficient, the crosswise force normalized by dynamic pressure and base area; C_m is the pitching-moment coefficient about the mass centre at 64% of length, normalized additionally by body length. Simulation across 0–3° of incidence gives $C_{N\alpha} = 2.04$ per radian and $C_{m\alpha} = -0.100$ per radian: the moment opposes any tilt, and the pressure centre sits 4.9% of the body length behind the mass centre, statically stable at the harshest point of the flight.

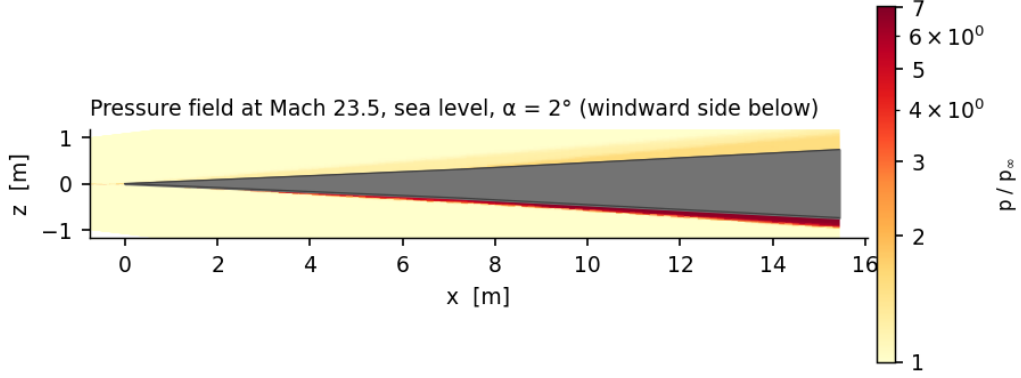


Figure 8: Pressure around the vehicle at Mach 23.5, sea level, 2° incidence, from the three-dimensional simulation; colors give pressure relative to the atmosphere.

These numbers come from three-dimensional simulations of the half-body on a structured surface-fitted mesh, with two stated simplifications: the surface is frictionless, because surface pressure on a slender body is set by flow turning while friction contributes drag, not stability; and the air does not dissociate, its temperature-varying heat capacities being included, and switching them off changes no coefficient before the fourth decimal, so the thermodynamic corrections are demonstrably small at these weak shock angles. The simulations were validated on both fronts available: an independent space-marching flow code reproduces the axial force to $+4.8\%$, and a coarser mesh moves the coefficients by only 0.2% .

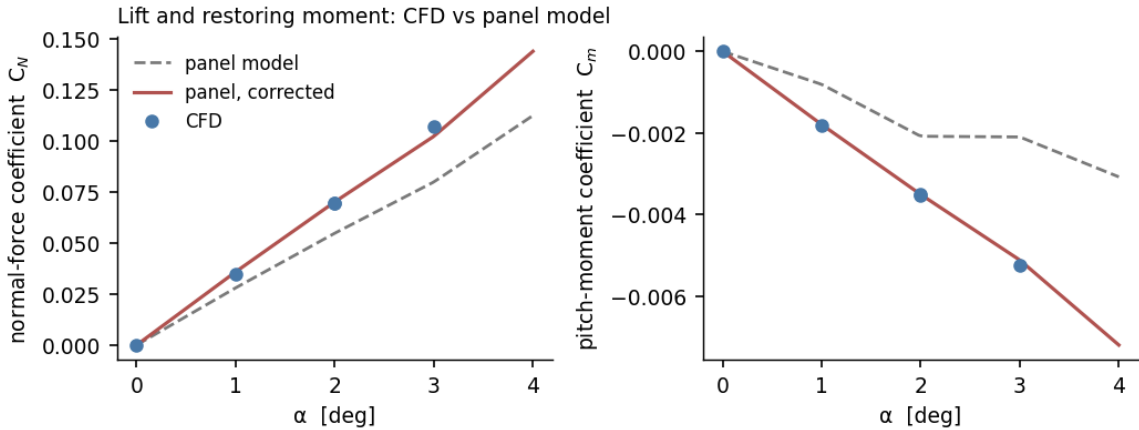


Figure 9: Normal-force and pitching-moment coefficients against incidence: simulation points, the simplified panel model, and the panel model scaled by a single factor.

The trajectory and dispersion studies of this program run on a simplified model that estimates the pressure on each patch of the surface from that patch's local inclination to the oncoming flow, ignoring how neighbouring patches disturb each other. This model was compared point by point against the simulations: it understates the normal force by a factor that is nearly the same at every incidence, 1.24 to 1.34 across the sweep. That uniformity permits a one-number repair, and the repair has been implemented: the panel normal force multiplied by 1.28, with the pressure centre placed at the simulated 69% of length, reproduces every simulation point within 5%, and the corrected tables now feed all downstream models. The uncorrected model erred in the safe direction for stability but understated trim drag and fin loads; those budgets shift accordingly.

Every payload number in this paper includes the full path to orbit. For the smaller vehicle the trajectory model, running on the corrected tables, charges 723 m/s to atmospheric drag, so the vehicle tops the atmosphere at 7.28 km/s; it then coasts on a ballistic arc to apogee, where the solid kick stage carried in its mass budget performs the circularization burn. The payload figures quoted throughout are the mass that survives this whole chain, drag, gravity, and burn included, not the mass at the muzzle.

8 Electricity is not the cost barrier

The marginal cost of a shot separates into energy, consumed hardware, and the vehicle:

USD per kg payload	16.3 km $\times$ 1.5 m	1000 km $\times$ 7 m	basis
drive electricity	3.8	2.5	simulated maps, 6 ¢/kWh
burst disks + initiator foils	3.5	0.7	5 ¢ each, one per port (assumed)
vehicle structure, motor, fuel	31.4	—	system model, expendable
marginal total, expendable	≈ 39	—	
floor: electricity + disks	7.3	3.2	

The electricity line passes the first-law audit it must: the small gun’s shot delivers 57.7 MWh of kinetic energy on 79 MWh from the plug, the difference leaving as compression heat and as the residual heat of the recovered gas; no regeneration between shots is credited. The disk line prices the part, not the operation: replacing ninety thousand port closures between shots is a turnaround burden that must be automated, and a closure that survives multiple shots is being sought; until one exists, the disk line carries an unpriced labor and cycle-time load beside it.

Capital sits outside the table, and the two designs face it differently. The small gun’s market makes its amortization easy to believe: below fifty dollars per kilogram, orbital solar power becomes buildable, and a single gigawatt-class plant in orbit is of order ten thousand tonnes of hardware, roughly eight thousand shots. Demand of that shape runs a gun continuously, and a fixed tube amortized over tens of thousands of shots adds little to the per-kilogram price. The 1000 km gun’s capital is of a different class, and its amortization is left untreated here; its case rests on the same logic applied to a civilization-scale traffic, which this paper asserts but does not price.

The per-port consumables matter because there are many ports: about 89,000 disks per shot on the small gun and 1.2 million on the large one, so the assumed nickel per stamped disk-and-foil assembly is itself a number the conclusion rests on. With it, the floor below which no vehicle improvement can push the price is \$7.30 per kilogram for the small gun and \$3.20 for the large one. The small gun’s expendable marginal total of about \$39 already sits under the fifty-dollar level of Section 6; it serves its market without vehicle recovery. Ten dollars is the large design’s target: its \$3.20 floor leaves \$6.80 for vehicle hardware, achievable through recovery or through mass production of a strictly identical vehicle. Barrel and plant are capital, excluded from the marginal figures.

9 Assumptions, and what could move the answer

The assumptions that carry the results, stated where they were made and collected here:

- all-steel gas system: wetted walls below 600 K, drive gas below 800 K;
- drive simulations are axisymmetric, laminar, single-ring-port; discrete ports and turbulence untested;
- ports pitched three widths apart with no jet interference;
- port force scales with bore area (checked to $\pm 0.3\%$);
- compression at 75% efficiency, electric heating at 95%, no heat regeneration credited; electricity at 6 ¢/kWh;
- barrel evacuated to ~ 100 Pa before each shot;
- hydrogen 99% recovered per shot; burst disks at 5 ¢ each;
- cone mass and its effect on vehicle balance not modeled (Section 5);
- vehicle aerodynamics inviscid, no dissociation;
- muzzle exit with zero tip-off and perfect trim, the friendliest case, though exit loads already size a third of the vehicle’s structure;
- no coupling of boundary-layer transition to surface roughness from ablation (a $3\text{--}5\times$ lever on heating);
- shell-buckling knockdown at its nominal value ($\pm 30\text{--}50\%$ on wall mass);
- standard atmosphere, no wind; payload modeled as load-bearing bulk at 800 kg/m^3 ; barrel erosion and refurbishment excluded from cost.

Two of these size real mass and deserve more than a line. The heat shield is sized by integrating the flank heat load along the flown trajectory at a fixed ablator effectiveness; if boundary-layer transition, seeded by the roughness of the ablating surface itself, multiplies flank heating threefold, the shield mass grows by roughly the same factor and comes out of payload, shrinking it without zeroing it. The exposure is that heating, recession, and trajectory are computed uncoupled, and coupling them is the first-ranked model improvement. The muzzle-exit disturbance is the structural twin of that problem: the side load of aerodynamic capture already sizes about a third of the vehicle’s structure with a perfectly aligned exit assumed, and a finite exit rotation raises that side load roughly in proportion, making the structure fraction the most exposed mass number in the paper; the planned resolution is a Monte Carlo over exit disturbances that sizes the wall at a stated percentile rather than at the friendliest case. The third mover, port-jet interference, would raise the pressure the electricity argument rests on if the three-width pitch proves too dense. None of the three threatens the architecture; each could move cost by tens of percent. Within these bounds the results support one claim: a fixed gun running cold hydrogen through wall ports can put mass in orbit for about ten dollars per kilogram.

Methods

Interior ballistics (Fig. 1). A one-dimensional unsteady finite-volume gas solver on a grid that stretches with the projectile, hydrogen with a high-density correction to the ideal gas law, wall friction, and injection ports triggered by projectile passage with feed-forward scheduling and pressure feedback. Injected gas enters at reservoir enthalpy with a selectable axial velocity, whose energy is debited from the gas’s enthalpy as a nozzle would. The solver reproduces the exact analytic expansion of a suddenly released gas column, and closes global mass and energy audits to numerical precision. Column curves use 128 ports spaced for equal transit time; the impingement curve is the design acceleration integrated directly, its feasibility resting on the maps of Fig. 4.

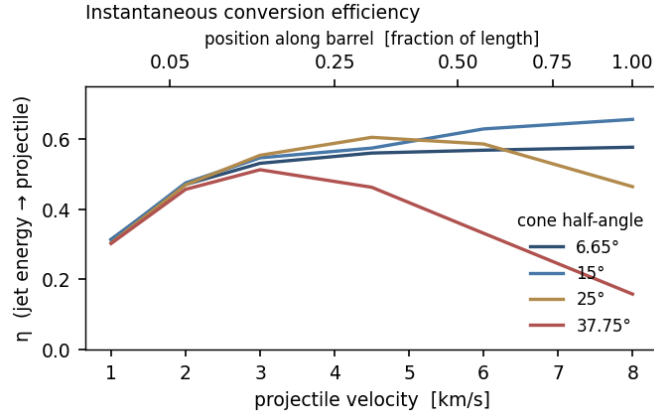


Figure 10: Instantaneous conversion efficiency against projectile velocity for four cone angles; the top axis gives the corresponding position along a constant-acceleration barrel.

Drive-flow simulations (Figs. 3–6). Axisymmetric compressible-flow simulations with a density-based shock-capturing solver from the OpenFOAM suite, hydrogen as a perfect gas, frictionless walls, laminar. The domain is a thin azimuthal wedge of the annular cavity between barrel wall and projectile: full-caliber forebody with the 3 mm clearance gap, boat-tail cone to a central stub, one port of 60 mm width in the wall, jet entering radially at 1500 m/s and 550 K in the barrel frame, transformed to the moving projectile frame at each flight speed. Thrust is the tail-averaged axial pressure force on cone and stub once the flow settles, between two and three cone-transit times. The campaign covers six velocities, seven cone angles, two port pressures, and repeats three cases at $1.6\times$ mesh density (thrust shifts of $+0.2$ to $+2.5\%$) and two at 7 m bore. The efficiency $\eta = Fv_p/\dot{m}(u_j^2/2 + c_p T_j)$ compares projectile power to delivered jet energy; its instantaneous form is shown above, its shot average in Fig. 6.

Energy ledger (Fig. 6). Hydrogen mass from integrating the inverse thrust map over velocity. Electricity per kilogram of gas is the sum of two terms: isothermal compression work at 300 K from the post-shot recovery pressure of the closed barrel to the required supply pressure, divided by 0.75, and electric heating from 300 K to the 628 K supply stagnation state, divided by 0.95. No heat regeneration

between shots is credited, because the recovered gas returns cold: its heat left with the projectile. The ledger closes the first law by construction, 79 MWh in against 57.7 MWh of kinetic energy out for the small gun; electricity priced at 6 ¢/kWh.

Vehicle aerodynamics (Figs. 7–9). Set-up and simplifications as given in Section 7; mesh of 240,000 cells, incidences of $0\text{--}3^\circ$, forces and moments taken about 64% of body length after the flow settles, air properties from standard thermochemical tables. The cross-validation code is an independent space-marching inviscid solver run on a published cone-flare geometry; the mesh check repeats one case at $0.72\times$ density.

Cost accounting. Structure and manufacturing at \$13/kg of structure, kick-stage fuel at \$6/kg, hydrogen make-up at \$6/kg on 1% loss per shot, burst-disk assemblies at 5 ¢ per port, electricity as above. Vehicle masses and payload fractions from the program’s system model at each design point; barrel, plant, and site are capital and excluded from marginal cost.